

Safety Data Sheet

Hydralan® 404

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(30133035/SDS_GEN_US/EN)

1. Identification

Product identifier used on the label

Hydralan® 404

Recommended use of the chemical and restriction on use

Recommended use*: brake fluid

Unsuitable for use: None known

* The "Recommended use" identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:

BASF CORPORATION
100 Park Avenue
Florham Park, NJ 07932, USA
Fuel and Lubricant Solutions

E-mail address: RegXcellenceFuelLubes@basf.com

Emergency telephone number

24 Hour Emergency Response Information

CHEMTREC: 1-800-424-9300

BASF HOTLINE: 1-800-832-HELP (4357)

Other means of identification

Chemical family: Blend based on: polyglycol, glycol ether, inhibitors

2. Hazards Identification

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

Classification of the product

Eye Dam./Irrit.	1	Serious eye damage/eye irritation
Repr.	2 (fertility)	Reproductive toxicity
Repr.	2 (unborn child)	Reproductive toxicity

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Label elements

Pictogram:



Signal Word:
Danger

Hazard Statement:

H318 Causes serious eye damage.
H361 Suspected of damaging fertility. Suspected of damaging the unborn child.

Precautionary Statements (Prevention):

P280 Wear protective gloves, protective clothing and eye protection or face protection.
P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.

Precautionary Statements (Response):

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310 Immediately call a POISON CENTER or physician.
P308 + P313 IF exposed or concerned: Get medical attention.

Precautionary Statements (Storage):

P405 Store locked up.

Precautionary Statements (Disposal):

P501 Dispose of contents/container in accordance with local regulations.

Hazards not otherwise classified

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

3. Composition / Information on Ingredients

According to Regulation 2012 OSHA Hazard Communication Standard; 29 CFR Part 1910.1200

Ethanol, 2-[2-(2-methoxyethoxy)ethoxy]-, triester with boric acid (H3BO3)

CAS Number: 30989-05-0

Content (W/W): ≥ 50.0 - $< 75.0\%$

Synonym: Ethanol, 2-[2-(2-methoxyethoxy)ethyl]-, triester with boric acid (H3BO3)

2-(2-(2-butoxyethoxy)ethoxy)ethanol

CAS Number: 143-22-6

Content (W/W): ≥ 3.0 - $< 5.0\%$

Synonym: 2-[2-(2-Butoxyethoxy)ethoxy]ethanol; Triethylene glycol monobutyl ether

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2-(2-methoxyethoxy)ethanol

CAS Number: 111-77-3

Content (W/W): ≥ 1.0 - $< 3.0\%$

Synonym: 2-(2-Methoxyethoxy)ethanol; Methyl carbitol

1,1'-iminodipropan-2-ol

CAS Number: 110-97-4

Content (W/W): ≥ 1.0 - $< 3.0\%$

Synonym: 1,1'-Iminobis(2-propanol); Diisopropanolamine

3,6,9,12-Tetraoxahexadecan-1-ol

CAS Number: 1559-34-8

Content (W/W): ≥ 0.3 - $< 3.0\%$

Synonym: No data available.

4. First-Aid Measures

Description of first aid measures

General advice:

Immediately remove contaminated clothing. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). First aid personnel should pay attention to their own safety.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention.

If on skin:

Remove contaminated clothing. Rinse skin immediately with plenty of water for 15 - 20 minutes. If irritation develops, seek medical attention.

If in eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

If swallowed:

Rinse mouth immediately with water. Never induce vomiting or give anything by mouth if the victim is unconscious or having convulsions. Do not induce vomiting due to aspiration hazard. Seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Information on: Ethanol, 2-[2-(2-methoxyethoxy)ethoxy]-, triester with boric acid (H₃BO₃)

Symptoms: Overexposure may cause:, nausea, headache, vomiting, dizziness, diarrhea, abdominal cramps

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Information on: 2-(2-(2-butoxyethoxy)ethoxy)ethanol

Symptoms: Overexposure may cause: corneal injury, skin corrosion, severe pain, coughing, respiratory disorders, dyspnea, nausea, headache, vomiting, dizziness, diarrhea, abdominal cramps

Information on: 1,1'-iminodipropan-2-ol

Symptoms: Overexposure may cause: abdominal cramps, dyspnea, coughing

Hazards: No applicable information available.

Indication of any immediate medical attention and special treatment needed

Note to physician

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
water spray, dry powder, foam

Special hazards arising from the substance or mixture

Hazards during fire-fighting:
harmful vapours

Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters

Protective equipment for fire-fighting:
Firefighters should be equipped with self-contained breathing apparatus and turn-out gear.

Further information:

The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Breathing protection required.

Environmental precautions

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up

For small amounts: Pick up with absorbent material (e.g. sand, sawdust, general-purpose binder).
Dispose of absorbed material in accordance with regulations.

For large amounts: Pump off product.

Place absorbed material in the same container as the spilled substance/product for disposal.

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7. Handling and Storage

Precautions for safe handling

Ensure thorough ventilation of stores and work areas.

Protection against fire and explosion:

Take precautionary measures against static discharges.

Conditions for safe storage, including any incompatibilities

No applicable information available.

Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

8. Exposure Controls/Personal Protection

No substance specific occupational exposure limits known.

Advice on system design:

No applicable information available.

Personal protective equipment

Respiratory protection:

Wear respiratory protection if ventilation is inadequate. Respiratory protection in case of vapour/aerosol release.

Hand protection:

Chemical resistant protective gloves

Eye protection:

Safety glasses with side-shields.

Body protection:

Body protection must be chosen based on level of activity and exposure., Protective coverall and/or impermeable apron and boots as necessary.

General safety and hygiene measures:

Wearing of closed work clothing is required additionally to the stated personal protection equipment. Keep away from food, drink and animal feeding stuffs. Avoid contact with skin and eyes. Remove contaminated clothing. No eating, drinking, smoking or tobacco use at the place of work.

9. Physical and Chemical Properties

Form:	liquid	
Odour:	of glycol	
Odour threshold:	No data available.	
Colour:	yellow	
pH value:	7 - 8.5	(FMVSS 116, S 6.4)
solidification temperature:	< -50 °C	(DIN ISO 3016)
Melting temperature:	No data available.	
Freezing point:	No data available.	

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Boiling point:	265 °C	(ASTM D1120)
Sublimation point:	No data available.	
Flash point:	135.5 °C	(ISO 2719)
Flammability:	hardly combustible	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling. The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Autoignition:	> 200 °C	(DIN EN 14522)
Vapour pressure:	1 mbar (20 °C) 1 mbar (50 °C)	
Density:	approx. 1.06 g/cm3 (20 °C)	
Relative density:	No data available.	
Vapour density:	> 1 (20 °C) Heavier than air.	(estimated)
Partitioning coefficient n-octanol/water (log Pow):	Study scientifically not justified.	
Self-ignition temperature:	not self-igniting	
Viscosity, dynamic:	not determined	
Viscosity, kinematic:	> 1.5 mm2/s (100 °C)	
Particle size:	The substance / product is marketed or used in a non solid or granular form.	
Solubility in water:	soluble	
Solubility (quantitative):	No data available.	
Solubility (qualitative):	soluble solvent(s): polar solvents,	
Molar mass:	No data available.	
Evaporation rate:	not determined	
Other Information:	If necessary, information on other physical and chemical parameters is indicated in this section.	

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals:

No corrosive effect on metal.

Oxidizing properties:

not fire-propagating

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

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Possibility of hazardous reactions

No hazardous reactions if stored and handled as prescribed/indicated.

Conditions to avoid

Avoid open flames.

Incompatible materials

strong oxidizing agents, atmospheric moisture

Hazardous decomposition products

Decomposition products:

Hazardous decomposition products: No hazardous decomposition products if stored and handled as prescribed/indicated.

2.5 K/min (DSC (DIN 51007))

11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Acute toxicity

Assessment of acute toxicity: Virtually nontoxic after a single ingestion. The product has not been tested. The statement has been derived from the properties of the individual components.

Oral

Type of value: ATE

Value: > 5,000 mg/kg

Inhalation

No applicable information available.

Dermal

No applicable information available.

Assessment other acute effects

Based on available data, the classification criteria are not met.

Irritation / corrosion

Assessment of irritating effects: May cause severe damage to the eyes. Not irritating to the skin.

Information on: 2-(2-(2-butoxyethoxy)ethoxy)ethanol

Assessment of irritating effects: Not irritating to the skin. Causes serious eye damage.

Eye

Result: Risk of serious damage to eyes.

The product has not been tested. The statement has been derived from the properties of the individual components.

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Sensitization

Assessment of sensitization: Based on the ingredients, there is no suspicion of a skin-sensitizing potential.

Aspiration Hazard

No aspiration hazard expected.

Chronic Toxicity/Effects

Repeated dose toxicity

Assessment of repeated dose toxicity: Based on available data, the classification criteria are not met.

Genetic toxicity

Assessment of mutagenicity: Based on available data, the classification criteria are not met.

Carcinogenicity

Assessment of carcinogenicity: None of the components in this product at concentrations greater than 0.1% are listed by IARC; NTP, OSHA or ACGIH as a carcinogen.

Reproductive toxicity

Assessment of reproduction toxicity: Based on the ingredients, there is a suspicion of a toxic effect on reproduction.

Information on: Ethanol, 2-[2-(2-methoxyethoxy)ethoxy]-, triester with boric acid (H3BO3)

Assessment of reproduction toxicity: The results of animal studies suggest a fertility impairing effect.

Teratogenicity

Assessment of teratogenicity: Possible risk of harm to the unborn child.

Information on: 2-(2-methoxyethoxy)ethanol

Assessment of teratogenicity: In animal studies the substance caused malformations.

Information on: Ethanol, 2-[2-(2-methoxyethoxy)ethoxy]-, triester with boric acid (H3BO3)

Assessment of teratogenicity: Indications of possible developmental toxicity/teratogenicity were seen in animal studies.

Other Information

The product has not been tested. The statements on toxicology have been derived from products of a similar structure and composition.

12. Ecological Information

Toxicity

Aquatic toxicity

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms.

Microorganisms/Effect on activated sludge

Toxicity to microorganisms

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Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations.

Mobility in soil

Assessment transport between environmental compartments

The substance will not evaporate into the atmosphere from the water surface.

Additional information

Other ecotoxicological advice:
Do not release untreated into natural waters.

The product has not been tested. The statements on ecotoxicology have been derived from products of a similar structure and composition.

13. Disposal considerations

Waste disposal of substance:

Dispose of in accordance with national, state and local regulations. It is the waste generator's responsibility to determine if a particular waste is hazardous under RCRA.

Container disposal:

Dispose of in a licensed facility. Recommend crushing, puncturing or other means to prevent unauthorized use of used containers.

14. Transport Information

Land transport

USDOT

Not classified as a dangerous good under transport regulations

Sea transport

IMDG

Not classified as a dangerous good under transport regulations

Air transport

IATA/ICAO

Not classified as a dangerous good under transport regulations

15. Regulatory Information

Federal Regulations

Registration status:

Chemical TSCA, US released / listed

EPCRA 311/312 (Hazard categories): Refer to SDS section 2 for GHS hazard classes applicable for this product.

EPCRA 313:

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<u>CAS Number</u>	Chemical name
111-77-3	2-(2-methoxyethoxy)ethanol
112-35-6	2-(2-(2-methoxyethoxy)ethoxy)ethanol
143-22-6	2-(2-(2-butoxyethoxy)ethoxy)ethanol

State regulations

<u>State RTK</u>	<u>CAS Number</u>	<u>Chemical name</u>
PA	110-97-4	1,1'-iminodipropyl-2-ol
	111-77-3	2-(2-methoxyethoxy)ethanol
	112-35-6	2-(2-(2-methoxyethoxy)ethoxy)ethanol
	143-22-6	2-(2-(2-butoxyethoxy)ethoxy)ethanol

Safe Drinking Water & Toxic Enforcement Act, CA Prop. 65:

WARNING: This product can expose you to chemicals including FORMALDEHYDE (GAS), which is known to the State of California to cause cancer, and BISPENOL A (BPA), which is known to the State of California to cause birth defects or other reproductive harm. For more information, go to www.P65Warnings.ca.gov.

NFPA Hazard codes:

Health: 3 Fire: 1 Reactivity: 0 Special:

HMIS III rating

Health: 3^a Flammability: 1 Physical hazard: 0

16. Other Information

SDS Prepared by:

BASF NA Product Regulations

SDS Prepared on: 2024/01/12

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